

# PING fundamentals

exp023 · 13 May 2026

## Abstract

Strips PING down to its biophysical fundamentals and characterises it *in isolation from any task* — no training, no readout, no loss, just a free-running network driven by Poisson input. The  $E \rightarrow I \rightarrow E$  loop produces gamma at  $\approx 30$  Hz with the standard  $\tau_{\text{AMPA}} + \tau_{\text{GABA}}$  cycle period, and the f-I curves show the loop is dynamic-range compression: PING holds E an order of magnitude below COBA across two orders of magnitude of drive, while COBA saturates near its refractory ceiling.

## Method

The architecture characterised here is sketched atop Figure 1. An input layer drives the excitatory population through  $W_{\text{in}}$ ; the  $E \rightarrow I \rightarrow E$  loop is formed by  $W_{\text{ei}}$  (E to I) and  $W_{\text{ie}}$  (I back to E), with no I→I synapse. Disabling the I→E loop (*ei-strength 0*) recovers the COBA reference; engaging it (*ei-strength 1.5*) produces the gamma rhythm.

The full conductance-based model that produces PING is documented in COBANet this notebook is the empirical companion. Two conditions run on the same network — *ei-strength 0* (loop disabled, the COBA reference) and *ei-strength 1.5* (loop engaged, the PING regime) — each driven by uniform Poisson input fed through  $W_{\text{in}}$  (untrained network). Because PING’s loop clamps the E rate while COBA’s is unclamped, a single input rate can’t show both legibly: the two rasters therefore use **per-condition input rates** — COBA at 5 Hz (a legible  $\approx 24$  Hz async raster), PING at 45 Hz (into gamma,  $f_\gamma \approx 40$  Hz). Population spectra are Welch PSDs on the population-mean E trace (one window per trial of length  $T$ , parabolic-interpolated peak; same recipe as exp041 and exp049).

The run scale reports the geometry that actually produced the figures. The two loop conditions (COBA, *ei-strength 0*; PING, *ei-strength 1.5*) are the grid; the raster/PSD input rates are per-condition (COBA 5 Hz, PING 45 Hz) and the free-running f-I panels sweep uniform Poisson drive (2–100 Hz,  $T = 400$  ms).

**Rate response.** The f-I panels of Figure 1 sweep spatially **uniform** Poisson input (all channels at the same rate, no digit structure) on the *untrained, free-running* network, comparing COBA against PING on a shared rate axis — the bare dynamic-range compression of the architecture.

## Results

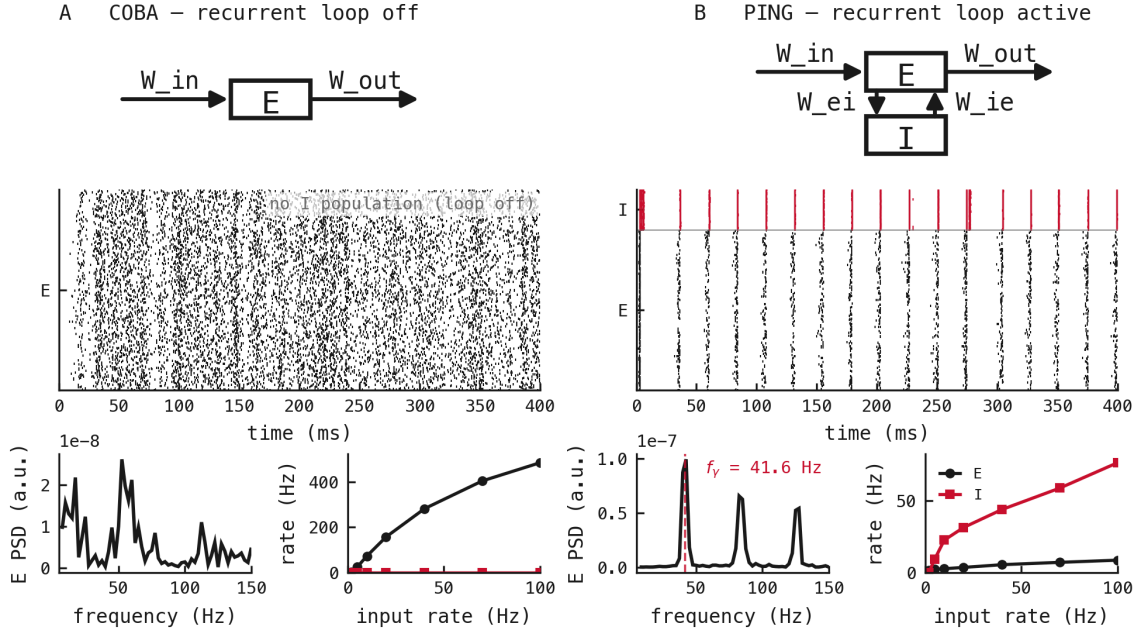


Figure 1: The architecture and its behaviour, off and on, on the same network. Each column carries its schematic on top — **A** COBA (input→E→output, no I population) and **B** PING (the same plus the recurrent E↔I loop,  $W_{ei}$  down and  $W_{ie}$  up) — directly above the plots it produces. Below each schematic is one combined raster (E black in the lower portion, I red in the upper portion — same axes, no gap) above a bottom row pairing the Welch PSD of the population-mean E trace with the free-running f-I curve (mean per-cell rate vs input rate, E black / I red; each column has its own y-axis — COBA runs to its  $\approx 400+$  Hz ceiling, PING is on a smaller scale so its loop-clamped E and climbing I are legible). Rasters and spectra are on uniform Poisson input (400 ms) at per-condition rates (COBA 5 Hz, PING 45 Hz — see Methods); the f-I curves sweep that drive on the untrained network. All runs are at  $\Delta t = 0.1$  ms. **A** — **COBA** (*ei-strength* 0, loop off): asynchronous E firing, no I activity, no recurrent gamma peak, and an E f-I curve that climbs steeply with input rate and saturates toward the refractory ceiling. **B** — **PING** (*ei-strength* 1.5, loop on): the I population fires a synchronous burst once per cycle and the E raster breaks into vertical bands (the I volley trails each E volley by the AMPA delay); the PSD carries a clean gamma peak (dashed red line;  $f_\gamma$  measured from this raster’s E-population PSD, marked on the plot); and its f-I shows the E rate held to single digits across the whole sweep (note the much smaller y-axis than COBA’s) while inhibition climbs — the loop’s dynamic-range compression. The single E→I→E loop is the only change: it converts asynchronous firing into a gamma rhythm *and* clamps the E rate.

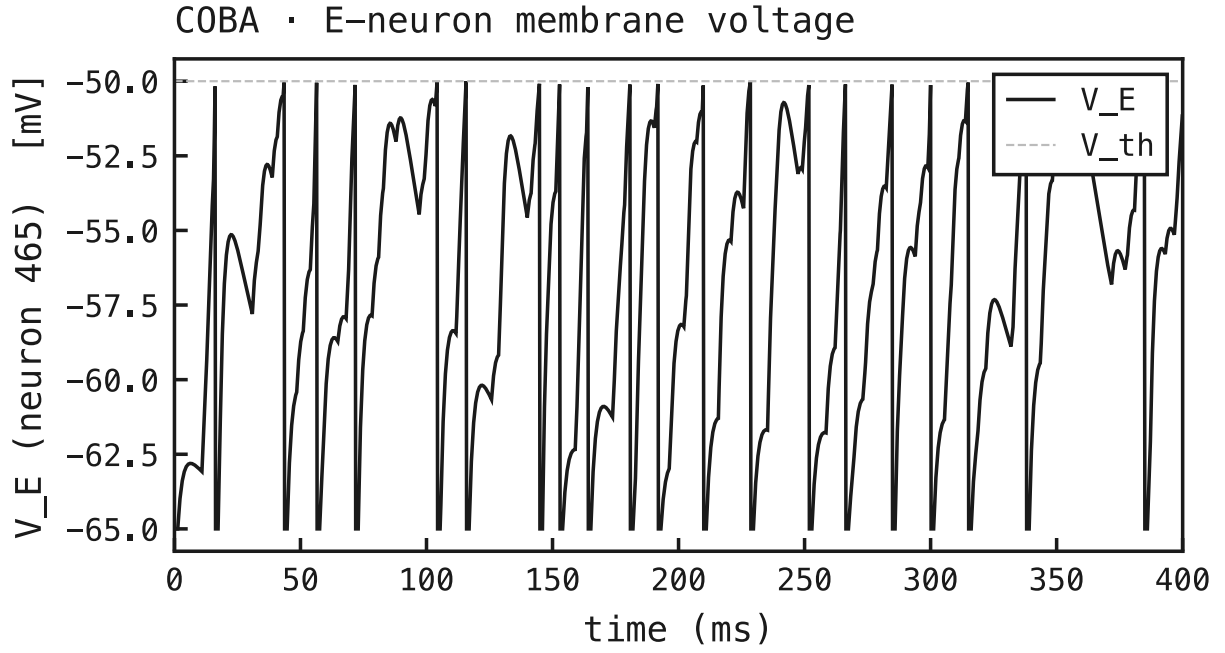


Figure 2: Membrane voltage  $V_E$  (black) of the most active E cell, recurrent loop **off** (COBA, *ei-strength* 0). Driven only by feedforward input, the cell charges toward threshold ( $V_{th} = -50$  mV, faint dashed) and hard-resets to  $V_{reset} = -65$  mV each time it fires — irregular, input-paced spiking with no rhythmic structure. The rhythm-free control for Figure 3a.

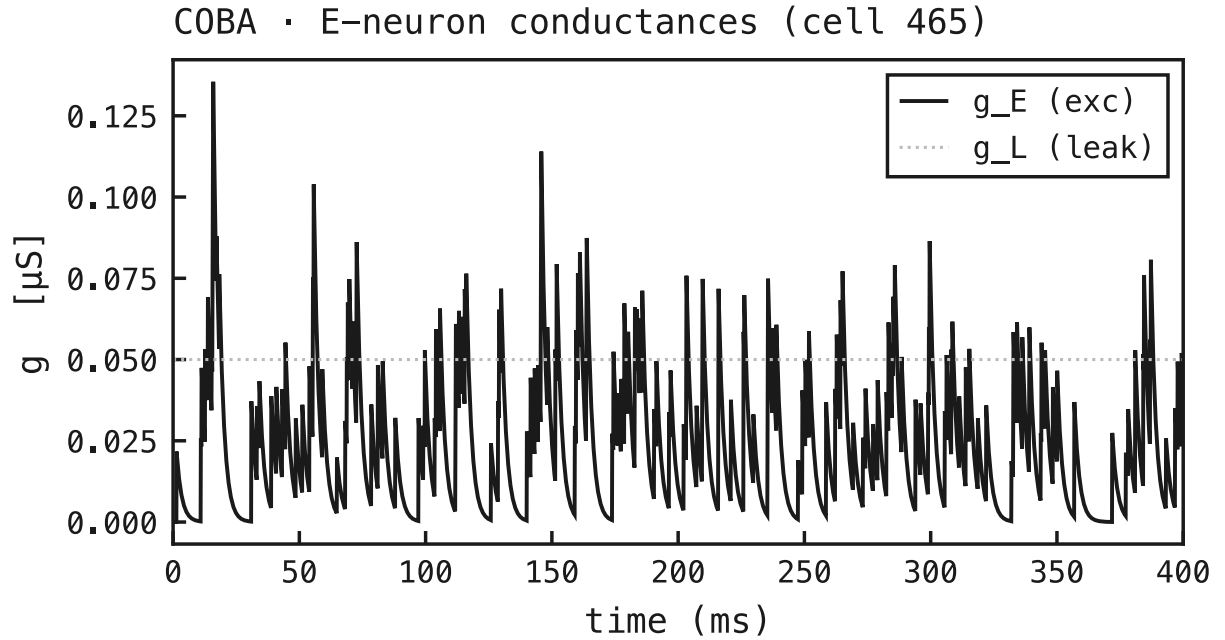


Figure 3: Synaptic and leak conductances on the same E cell, COBA mode. Only the excitatory conductance  $g_E$  (black) is active — it steps up with each input spike and decays with  $\tau_{AMPA}$  — while the leak  $g_L$  (faint dotted) is fixed. There is no inhibitory  $g_I$  because the  $I \rightarrow E$  loop is off. All conductances are **non-negative**: they count open channels, not current direction or sign.

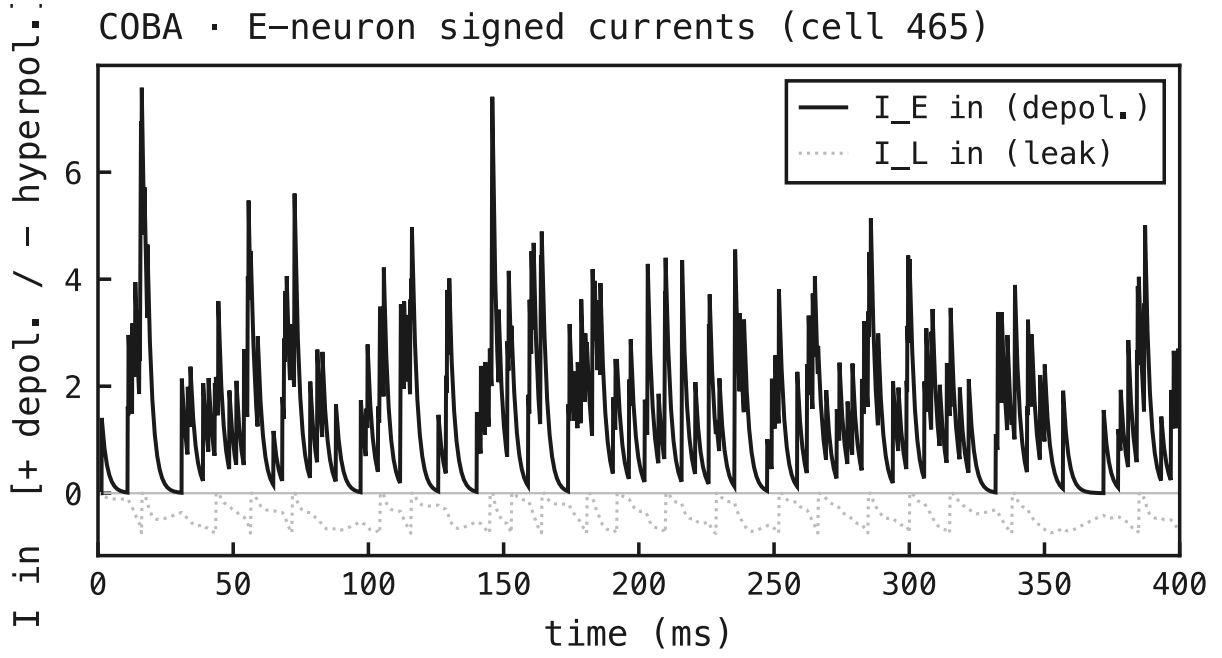


Figure 4: Signed synaptic currents into the E cell, COBA mode, with  $I_X^{\text{in}} = -g_X(V - E_X)$  (positive = depolarising). With no inhibition, the only synaptic current is the depolarising excitatory current  $I_E^{\text{in}}$  (black); the leak current (faint) hovers near zero. Compare Figure 3c, where the loop introduces an inhibitory current of the *opposite* sign.

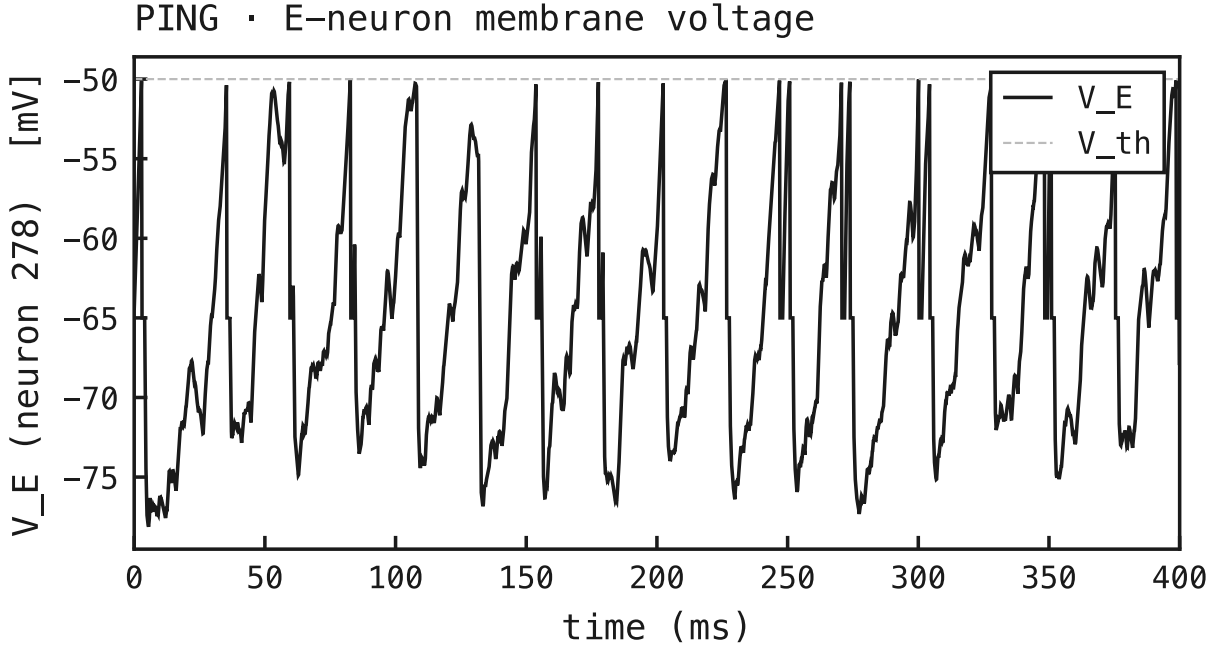


Figure 5: Membrane voltage  $V_E$  (black) of the most active E cell, recurrent loop **on** (PING, *ei-strength* 1.5), same input as 2a. Rhythmic inhibitory bursts now hold the cell below threshold most of the time: each I-burst drags  $V_E$  toward  $E_i = -80$  mV, and the membrane recovers — and can fire — only as the inhibition decays between bursts. The loop has turned the irregular firing of 2a into cycle-gated firing.

### PING · E-neuron conductances (cell 278)

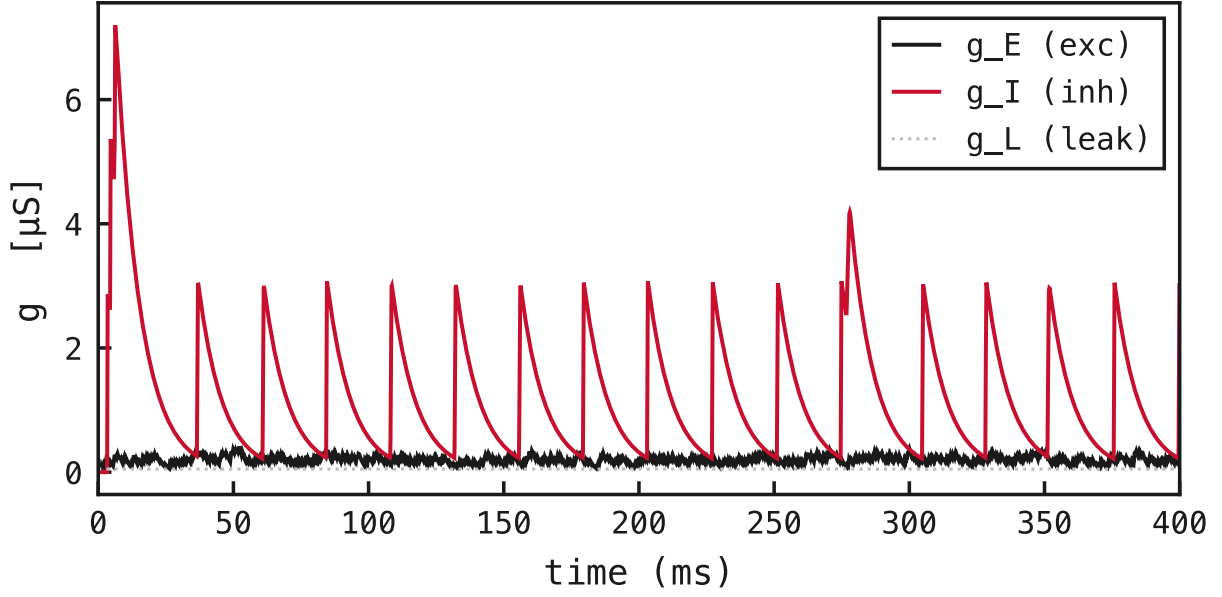


Figure 6: Conductances on the same E cell, PING mode. The inhibitory conductance  $g_I$  (red) now dominates — it spikes once per gamma cycle as the synchronous I-burst arrives through  $W_{ie}$ , then decays with  $\tau_{GABA}$  — while  $g_E$  (black) carries the smaller feedforward input and  $g_L$  (faint) is fixed. All three traces are **non-negative**:  $g_I$  is large and positive, and that is what shunts the cell — it carries no sign of its own.

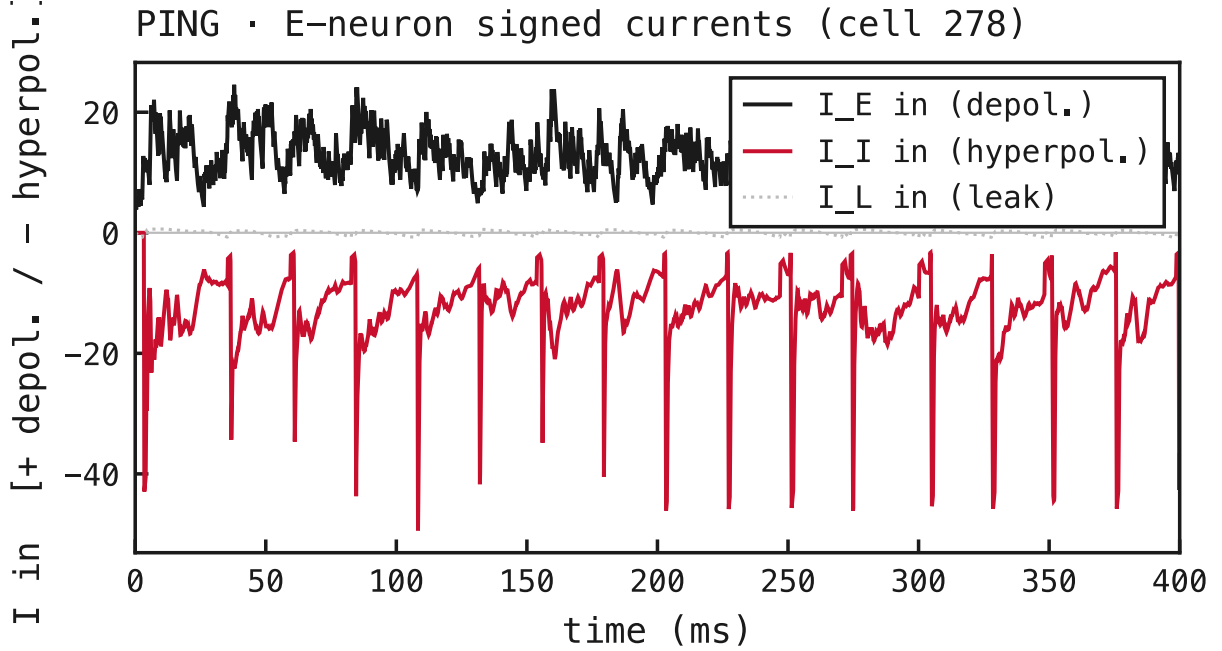


Figure 7: The load-bearing panel. Signed currents into the E cell, PING mode. The excitatory current  $I_E^{\text{in}} = -g_E(V - E_E)$  (black) is depolarising; the inhibitory current  $I_I^{\text{in}} = -g_I(V - E_I)$  (red) is **negative (hyperpolarising) even though  $g_I$  is positive** (Figure 3b) — the minus sign comes entirely from the driving force ( $V - E_I > 0$  whenever  $V$  sits above  $E_i = -80$  mV). The COBA principle made literal: *the sign lives in the driving force, never in the conductance*. The sharp negative pulses are the per-cycle inhibitory kicks that gate the rhythm.

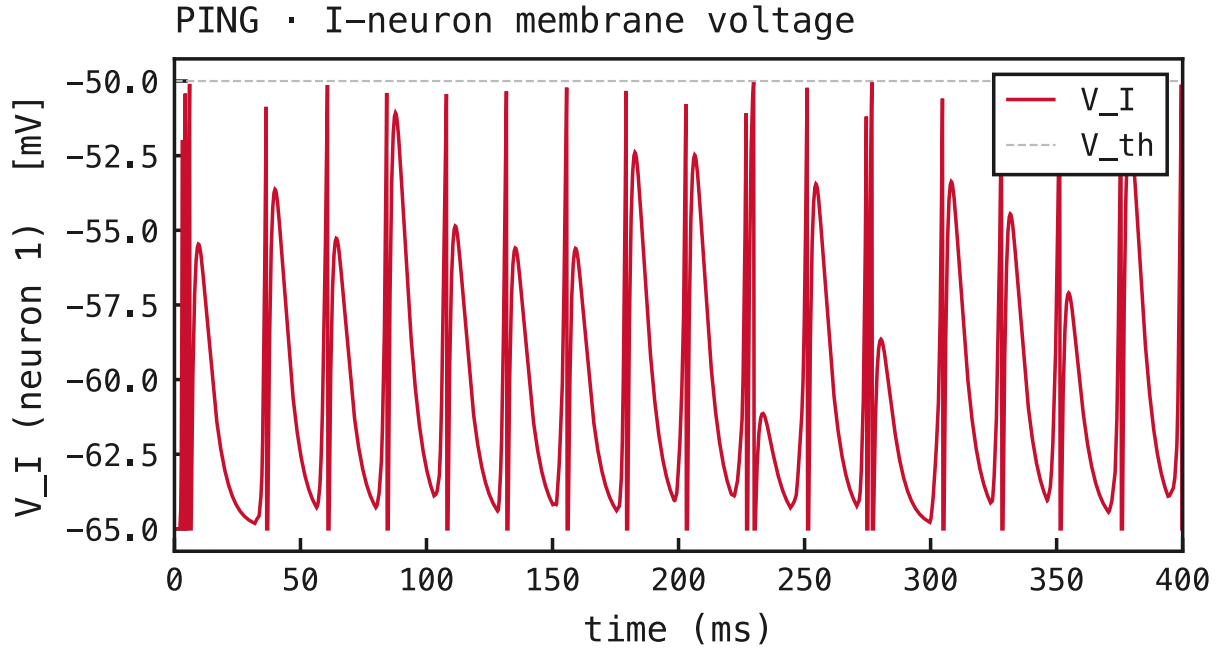


Figure 8: Membrane voltage  $V_I$  (red) of the most active I cell, PING mode. The I cell integrates the excitatory volley from the E population and fires once per gamma cycle, so  $V_I$  tracks the E-burst envelope — ramping up as E cells fire, crossing threshold ( $V_{th} = -50$  mV, faint dashed), resetting, and waiting for the next volley. This single I-spike per cycle delivers the inhibitory burst seen on the E cell in 3b–3c.

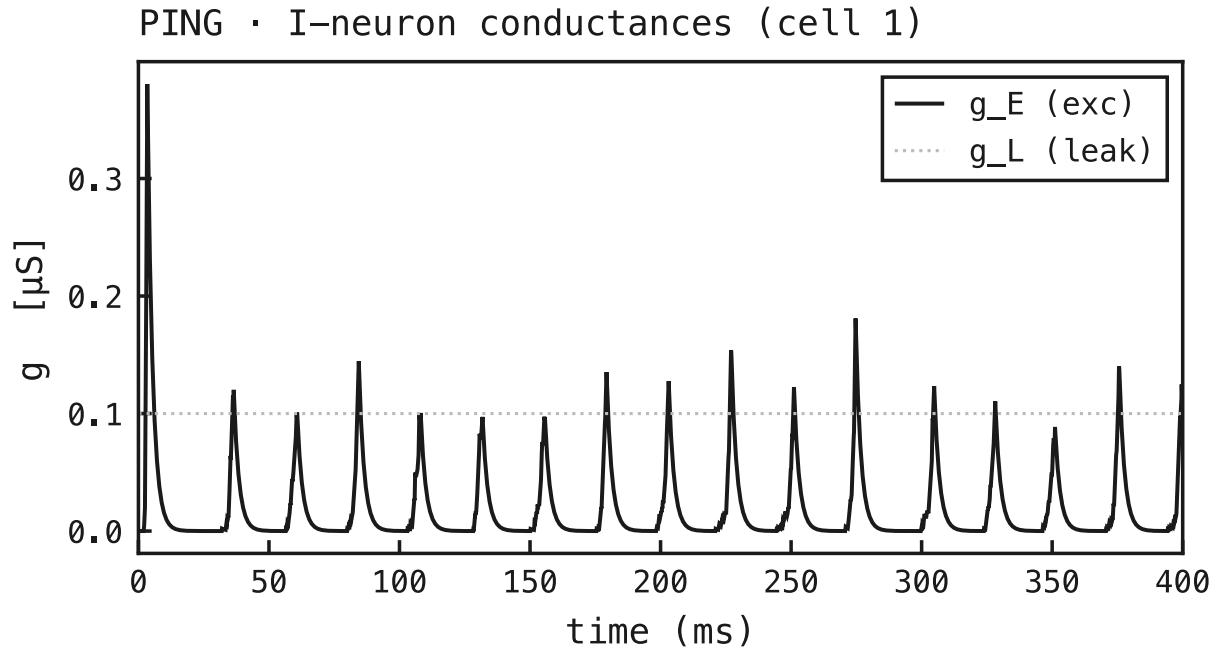


Figure 9: Conductances on the I cell, PING mode. The I cell receives **only excitation**:  $g_E$  (black) is the arriving E-spike envelope (delivered via  $W_{ei}$ ) and  $g_L$  (faint) is the fixed leak. There is no inhibitory conductance — the architecture has no I→I synapse — which is exactly why the I population can synchronise into the sharp once-per-cycle bursts that clock the loop.

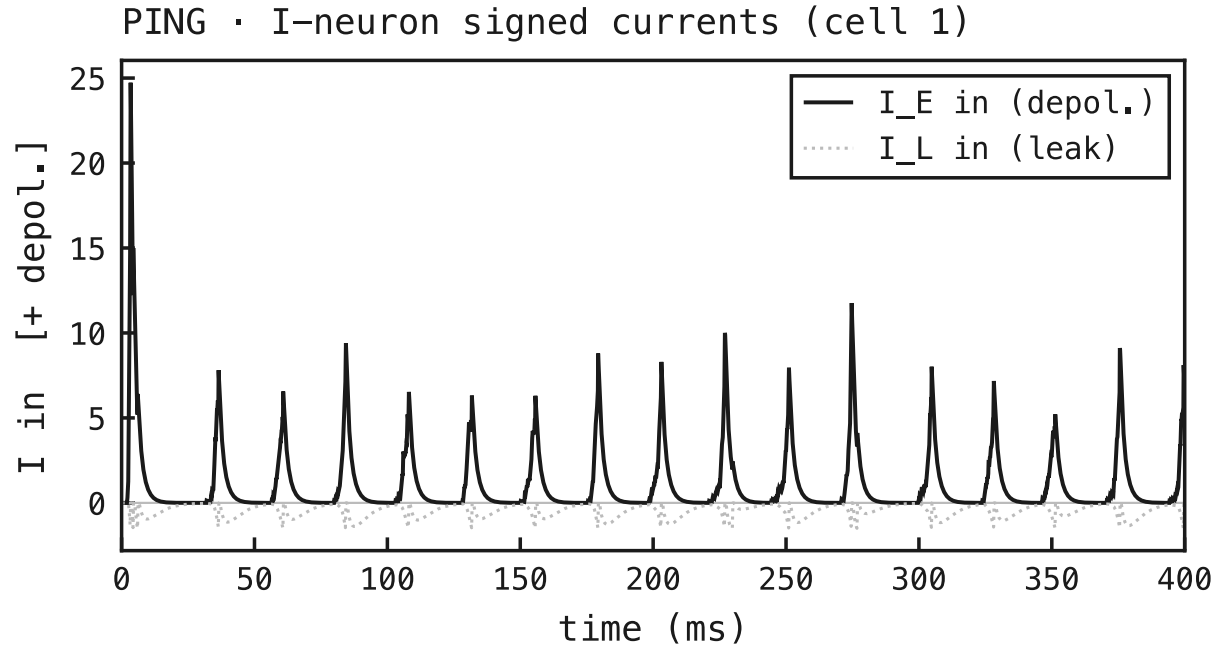


Figure 10: Signed currents into the I cell, PING mode. With no inhibitory input, only the depolarising excitatory current  $I_E^{\text{in}}$  (black) and a small leak current contribute — the I cell is driven purely up to threshold. Read 3a–3f together and the loop is in cross-section: E spikes ramp  $g_E$  on I (3e–3f) → I fires once per cycle (3d) → that spike delivers the inhibitory burst  $g_I$  on E (3b) → its negative current shuts E down (3c, 3a) →  $g_I$  decays → E refires. That delayed E→I→E loop is PING.